

# Umurima AI

Farming advice over SMS for smallholder farmers in Rwanda who use basic phones

Hack-Nation × World Bank · Small AI for Development · Challenge 04, Agriculture · October 2026  
Code: [github.com/Dawit-Getachew/umurima](https://github.com/Dawit-Getachew/umurima) · Live API: [api.aisms.coolify.appilab.com/docs](https://api.aisms.coolify.appilab.com/docs) · Team: CMU-Africa bootcamp, Group 7

## Summary

A farmer texts a question, in Kinyarwanda or English, to an ordinary phone number. An Android gateway phone relays it to the Umurima backend, which replies within seconds with one concrete recommendation and the person to ask next, in at most two SMS. The farmer needs no smartphone, no data bundle and no app. Emergencies, such as a sick animal or pesticide poisoning, are handled by fixed rules that send the farmer to a vet or a health centre.

## Problem statement

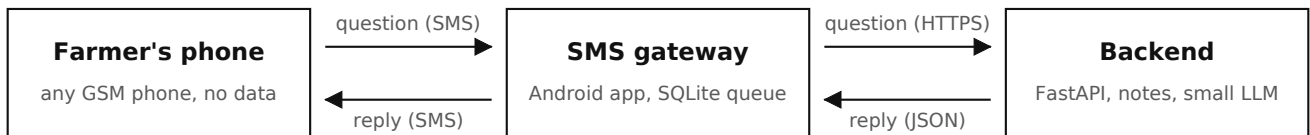
Because of Umurima AI, a smallholder farmer like Noor will get a specific answer to a crop or animal question within a minute, by SMS on the phone she already has, that she would otherwise wait months for, since the extension officer reached her village twice last year. We know because 78% of Rwandan agricultural households have a mobile phone but only 9% a smartphone, and 32% of household heads cannot read or write (NISR, Census 2022).

## Why SMS

| Figure                                                                                     | Source                                      |
|--------------------------------------------------------------------------------------------|---------------------------------------------|
| 69% of Rwandan households (2.3 million) are agricultural                                   | NISR, Census 2022                           |
| In agricultural households: 78% have a mobile phone, 9% a smartphone, 16% internet access  | NISR, Census 2022                           |
| 62% of workers are in agriculture; 34% of households own a smartphone (23% in rural areas) | NISR, EICV7 2023/24                         |
| Smartphones are 22% of mobile connections; mobile internet usage gap of 80%                | GSMA, 2024                                  |
| Average farm is 0.4 ha; 77% of agricultural households farm under 0.5 ha                   | NISR, Agricultural Household Survey 2019/20 |
| 14,000+ Farmer Promoters trained under Twigire Muhinzi                                     | RAB, 2025                                   |

The constraints follow from these figures: a basic phone, no data connection, limited literacy, and a human extension network that exists but is spread thin. Umurima works with that network rather than around it: every reply names a Farmer Promoter, agronomist or vet.

## How it works



Only questions not answered by rules, verified answers or the cache reach the language model.

The backend answers each message through five layers, ordered by cost and risk. Only the fourth needs the language model.

| Layer               | What it does                                                                                                                                                                                                                   |
|---------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1. Triage           | Rules detect emergencies. A sick or dying animal and a spreading crop problem get two SMS: a safe step to take now, then a fixed escalation to the vet or agronomist. A poisoned person gets one fixed SMS: health centre now. |
| 2. Verified answers | 27 reviewed answers to the most common questions, returned instantly and identically every time.                                                                                                                               |
| 3. Answer cache     | Earlier answers matched on the question's meaning rather than its exact words. Persisted, and retired automatically when the knowledge or prompt changes.                                                                      |
| 4. Retrieval + LLM  | BM25 search over agronomy notes written from RAB, MINAGRI and NAEB guidance, plus the last six messages, sent to Gemini 3.1 Flash Lite. Kinyarwanda farm words are mapped to the notes' terms.                                 |
| 5. Offline fallback | If the model is unreachable: the best-matching sentence from the notes, or an honest "try again" with a person to ask.                                                                                                         |

## Real replies from the live service

| Farmer sends                                                                                             | Umurima replies                                                                                                                                                                                                                                                                                                                                              |
|----------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| My maize leaves have holes and sawdust in the funnel                                                     | Your maize is likely attacked by fall armyworm. Scout your field twice a week and remove the larvae by hand while they are still small to prevent further damage. More help: your Farmer Promoter or sector agronomist.                                                                                                                                      |
| Ibigori byanjye bifite amababi y'umuhondo, nkore iki?<br>(My maize leaves are yellow, what should I do?) | Ibigori byawe bishobora kuba bibuze ifumbire ya azote. Ongerera ifumbire ya urea ku butaka buhehereye igihe ibigori bigize amavi, kandi ushyiremo ifumbire y'imborera cyangwa iy'ikimoteri. Ubundi bufasha: umuhinzi mwitozwa cyangwa agronome w'umurenge.<br>(Your maize may lack nitrogen. Add urea on moist soil at knee height, plus manure or compost.) |
| My cow is not eating and has fever                                                                       | (1/2) Your cow may be suffering from a tick-borne disease. Keep the animal in a cool, shaded area and provide plenty of clean water while you wait for help.<br>(2/2) This is urgent. Call your sector veterinary officer today. Do not give old or leftover drugs, and do not eat or sell meat from an animal that died sick.                               |

## Why AI, and not a simpler tool

An SMS menu cannot understand a free-text question such as “ibigori byanjye bifite amababi y'umuhondo”. A spreadsheet cannot be searched from a button phone, and a web search needs a smartphone, data and English. The AI layer understands the question in either language, finds the right vetted note, and writes an answer that fits in two SMS. It diagnoses from a description (holes and sawdust in the funnel: fall armyworm) instead of matching keywords.

Most traffic never reaches the model. Rules, verified answers and the cache answer instantly and at no cost, and they keep working when the model is unreachable.

## Responsible AI

- **A person makes the decision.** The tool only advises. Every reply ends with a human contact appended by code, not generated: Farmer Promoter, sector agronomist or vet.
- **Escalation by rules, not by the model.** A dying animal or a poisoned person always reaches a person. The first SMS for an animal is checked in code; any drug, dose, injection or number replaces it with a fixed safe step. Human first aid is never generated.
- **No invented numbers.** No doses, prices, dates or phone numbers that are not in the notes. Dose questions get the practical step plus “use the rate on the bag”.
- **Honest when unsure.** “I am not sure” plus a person to ask, instead of a guess. Off-topic questions are declined.
- **Privacy.** Only phone number, message and reply are stored, to give the last six turns as context. The phone number is never sent to the model provider. The gateway can receive and send SMS but cannot read the phone's inbox.

## Small AI fit

| Constraint                        | How Umurima meets it                                                                                                    |
|-----------------------------------|-------------------------------------------------------------------------------------------------------------------------|
| Device the user already has       | Any GSM phone; SMS works on 2G with no data.                                                                            |
| Weak or intermittent connectivity | The gateway stores every SMS before upload and retries until it is answered.                                            |
| Small footprint                   | About 20 KB of knowledge text and an in-memory BM25 index; rules, verified answers and the cache run without the model. |
| Local language                    | Kinyarwanda and English, detected per message; emergency replies are hand-written in both.                              |
| Cost and speed                    | Measured: about US\$0.0002 per AI answer and 1 to 3 seconds; verified and cached answers in milliseconds.               |

## Data the tool is built with

No model was trained or fine-tuned, and no synthetic data was used. The tool is grounded in eight knowledge notes (68 paragraphs, about 20 KB) written by the team from public RAB, MINAGRI and NAEB guidance; 27 verified answers derived from them; fixed emergency replies in two languages; and a map of 64 Kinyarwanda farm terms.

### What the data does not cover

- No review by RAB yet: the notes are drafts written from public material.
- No plot-level soil, rainfall or satellite data; advice is national or regional.
- No fertilizer or pesticide rates, live market prices or weather forecasts.
- Thin coverage of horticulture, pigs, rabbits, fish and bees.
- Kinyarwanda output is weaker than English; dialects and SMS spelling are untested.
- No usage data from real farmers yet.

## Technology

| Part        | Stack                                                                                                 |
|-------------|-------------------------------------------------------------------------------------------------------|
| SMS gateway | Flutter (Android), SQLite queue, foreground service, telephony plugin                                 |
| Backend     | Python, FastAPI, rank-bm25, OpenRouter (Gemini 3.1 Flash Lite), Postgres or SQLite, Docker on Coolify |
| Quality     | 34 backend tests and 7 app tests; every demo question verified against the live model                 |

## Next steps

- Review of the notes and verified answers by RAB agronomists, and of the Kinyarwanda output by a native-speaking agronomist.
- Consent message on first contact, a STOP command, authentication and rate limiting.
- A telco short code, so a single gateway phone is no longer a point of failure.
- A Kinyarwanda voice line for farmers who cannot read SMS.

## What localizing AI means here

Localizing AI means starting from the farmer rather than the model: the phone she already owns, her language, the extension workers she already trusts, and national guidance she can act on today. It also means being honest about the limits, and keeping a person in charge of every decision that matters.

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Code and documentation: [github.com/Dawit-Getachew/umurima](https://github.com/Dawit-Getachew/umurima) · Live API: [api.aisms.coolify.appilab.com/docs](https://api.aisms.coolify.appilab.com/docs) · License: MIT